Catalysts

RAJIB
Che-50

2019 (t), 2020, 2016

⑧ What is catalysis and a catalyst? What are different types of catalysis? Give one example of each type in solid, liquid and gas phase?

⇒ A catalyst is defined as a substance which alters the rate of a chemical reaction (rxn), itself remaining chemically unchanged at the end of the reaction. And The process is called catalysis. A catalyst may increase or decrease the rate of a reaction. A catalyst which enhances the rate of a reaction is called a positive catalyst and the process is positive catalysis. A catalyst which retards the rate of a reaction is called negative catalyst and the process is called negative catalysis.

⇒ There are two main types of catalysis: ① Homogeneous ② Heterogeneous catalysis.

Also, there is a third type of catalysis known as

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enzyme catalysis.

⇒ Heterogeneous catalysis are chemical reactions in which the reactants & ~~products~~ catalyst are not in the same phase. The most important of such reaction are those in which the reactants are in the gas phase while the catalyst is a solid. The process is also called contact catalysis since the reaction occurs by contact of reactants with the catalyst surface.

Ex: ① Heterogeneous catalysis with gaseous reactants:

(a) combination of SO_2 and O_2 in presence of finely divided platinum or V_2O_5,

$$2SO_2(g) + O_2(g) \xrightarrow{Pt(s)/V_2O_5(s)} 2SO_3(g)$$

(b) combination of N_2 and H_2 to form NH_3 in the presence of finely divided iron,

$$N_2(g) + 3H_2(g) \xrightarrow{Fe(s)} 2NH_3(g)$$

② Heterogeneous catalysis with liquid reactants:

(a) $2H_2O_2(l) \xrightarrow{MnO_2(s)/Pt(s)} 2H_2O(l) + O_2(g)$

(b) $C_6H_6(l) + CH_3COCl(l) \xrightarrow{AlCl_3(s)} C_6H_5COCH_3 + HCl$

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③ Heterogeneous catalysis with solid reactants

$$2KClO_3(s) \xrightarrow{MnO_2} 2KCl + 3O_2$$

EX. ① Homogeneous catalysis in gas phase

(a) $2SO_2(g) + O_2(g) \xrightarrow{NO(g)} 2SO_3(g)$

② Homogeneous catalysis in soln phase (liquid phase):

$$C_{12}H_{22}O_{11}(l) + H_2O(l) \xrightarrow{H^+(l)} C_6H_{12}O_6 + C_6H_{12}O_6$$

Homogeneous c. refers to the reactions where the catalyst is in the same phase as the reactants. This type can occur in gas or liquid phas

2022(t), 2021(t), 2015, 2016, 2018,

⊗ Explain the characteristics of catalyst ?

⇒ Characteristics of catalysts are:

1. The catalyst remain unchanged in mass and chemical composition: Catalyst may undergo a physical change but it undergoes no change in mass of chemical nature.
2. Small amount of catalyst is needed: sometimes a trace of a metal catalyst is required to affect very large amounts of reactants of reactants.
3. Effect of the equilibrium of a reversible reaction: It implies that in a reversible reaction the catalyst accelerates the forward and the reverse reaction

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equally. For example, In haber process for NH_3, the Iron catalyst shortens the time to attain equilibrium but cannot alter the percentage yield.

4. Inability to start a reaction: In most cases a catalyst speeds up a reaction already in progress and does not initiate (or start) the reaction.

5. Specific nature of a catalyst: While a particular catalyst works for one reaction, it will no necessary to work for another reaction. Different catalysts can bring about completely different reactions for the same substance.

6. A catalyst is more effective when: finely divided: In heterogeneous catalysis, the solid catalyst is more effective when in a state of fine subdivision than it is used in bulk. For example finely divided nickel is a better catalyst than lumps of solid nicked..

7. Nature of the products is unaltered (→ পরিবর্তন) by the presence of the catalyst: Most cases it is true but this is

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not always true. There are some exceptions. General example are N_2 & H_2 always give NH_3 in presence or absence of catalyst. similarly SO_2 & O_2 always give SO_3 in presence or absent of catalyst.

8. Optimum temperature: Some catalysts are, physically altered by a rise in temperature & hence their catalytic activity may be decreased.

9. Action of promoters: The activity of a catalyst can often be increased by addition of a small quantity of a second material.

10. A catalyst is poisoned by certain substance: Very often a heterogeneous catalyst is rendered ineffective by the presence of small amounts of impurities in the reactants.

★ 2015 What is autocatalysis?

⇒ When one of the products of the reaction itself acts as a catalyst for that reaction the pheriomenon is called auto catalysis. In auto catalysis the initial

rate of the reaction rises as the catalytic product is formed, instead of decreasing steadily. Example: The hydrolysis of ethyl acetate forms acetic acid and ethanol. of these products, acetic acid acts as a catalyst for the reaction.

$$CH_3COOC_2H_5 + H_2O \longrightarrow \underset{\text{(catalyst)}}{CH_3COOH} + C_2H_5OH$$

2019(T), 2015, 2020, 2022, 2017

⊛ What is acid-base catalysis? what are different types of acid base catalysis? Discuss the mechanism & Kinetics of acid base catalysis?

⇒ A number of homogeneous catalytic reactions are known: which are catalysed by acids or bases, or both acids & bases. These are often referred to as acid-base catalysts. These phenomenon is called acid-base catalysis. Arrhenius pointed out that acid catalysis was, in fact, brought about by H^+ ions supplied by strong acids, while base catalysis was caused by OH^- ions supplied by strong bases. Example of acid-

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base catalysis:

1. Inversion of cane sugar:

$$C_{12}H_{22}O_{11} + H_2O \xrightarrow{H^+} C_6H_{12}O_6 + C_6H_{12}O_6$$

2. Keto-enol tautomerism of acetone:

$$CH_3COCH_3 \xrightarrow{H^+} CH_3C(OH)=CH_2$$

3. Hydrolysis of an ester:

$$CH_3COOC_2H_5 \xrightarrow{H^+} CH_3COOH + C_2H_5OH$$

There are two types: acid catalysis & base catalysis.

⇒Mechanism:

(a) In acid catalysis, the H^+ forms an intermediate complex with the reactant, which then reacts to give back the proton. For example, the mechanism of keto-enol tautomerism of acetone is:

$$CH_3\overset{O}{\overset{\|}{C}}-CH_3 \xrightarrow{H^+} CH_3-\overset{OH^+}{\overset{\|}{C}}-CH_3 \longrightarrow CH_3-C(OH)=CH_2 + H^+$$

(b) In base catalysis, the OH^- ion (or any Bronsted base) accepts a proton from the reactant to form an intermediate complex which then reacts or decomposes to regenerate the OH^-. For example, the decomposition of nitramide by OH^- ions and CH_3COO^- ions may be

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explained as follows:

(i) By OH^- ions:

$$NH_2NO_2 + OH^- \longrightarrow H_2O + NHNO_2^-$$

(intermediate complex)

$$NHNO_2^- \longrightarrow N_2O + OH^-$$

(ii) By CH_3COO^- ions:

$$NH_2NO_2 + CH_3COO^- \longrightarrow CH_3COOH + NHNO_2^-$$

$$NHNO_2^- \longrightarrow N_2O + OH^-$$

$$OH^- + CH_3COOH \longrightarrow H_2O + CH_3COO^-$$

⇒ Kinetics of acid-base catalysis:

Acid-base catalysts reactions take place by transfer of proton from an acid to a substrate molecule (s), or from a substrate molecule to a base. Hence the reaction btwn an acid HA & a substrate, s may be occured as follows:

$$HA + S \underset{K_{-1}}{\overset{K_1}{\rightleftharpoons}} SH^+ + A^-$$

$$SH^+ + H_2O \xrightarrow{K_2} \text{products}$$

Reaction rate $= K_2[SH^+]$

By applying steady-state principle,

$$d[SH^+]/dt = K_1[HA][S] - K_{-1}[SH^+][A^-] - K_2[SH^+] = 0$$

$$\Rightarrow [SH^+] = K_1[HA][S] / (K_{-1}[A^-] + K_2)$$

case-I: If $K_2 \gg K_{-1}[A^-]$ then,

$$[SH^+] = K_1[HA][S] / K_2$$

Therefore, reaction rate $= K_1[HA][S]$

case-II: If $K_2 \ll K_{-1}[A^-]$ then,

$$[SH^+] = K_1[HA][S] / K_{-1}[A^-]$$

Now consider the dissociation of HA

$$HA \rightleftharpoons H^+ + A^-$$

$$\therefore K_a = [H^+][A^-] / [HA]$$

$$\Rightarrow [HA]/[A^-] = [H^+]/K_a$$

Therefore, $[SH^+] = K_1[H^+][S] / K_a K_{-1}$ and

reaction rate $= K_1 K_2 [H^+][S] / K_a K_{-1}$

$= K'[H^+][S]$ [let, $K' = K_1 K_2 / K_a K_{-1}$]

Hence the reaction rate is proportional to $[H^+]$ even in the presence of HA and A^-.

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2015, 2021(I), 2016, 2021, 2022

(*) Explain Homo & Heterogeneous catalysis with example?

⇒ Homogeneous catalysis refers to the reactions where the catalyst is in the same phase as the reactants. This type of catalysis can occur in gas phase or the liquid phase.

Examples of homogeneous catalysis in gas phase:

(a) $2SO_2(g) + O_2 \xrightarrow{NO(g)} 2SO_3(g)$

(b) $CH_3CHO(vap) \xrightarrow{I_2(vap)} CH_4(g) + CO(g)$

Examples of homogeneous c. in soln. phase:

(a) $C_{12}H_{22}O_{11}(l) + H_2O(l) \xrightarrow{H^+(l)} C_6H_{12}O_6 + C_6H_{12}O_6$

(b) $CH_3COOCH_3(l) + H_2O(l) \xrightarrow{H^+(l)} CH_3COOH(l) + CH_3OH(l)$

(c) $2H_2O_2(l) \xrightarrow{I^-(l)} 2H_2O + O_2$

⇒ Heterogeneous catalysis are chemical reactions in which the reactants and ~~products~~ catalyst are not in the same phase.

Examples of heterogeneous catalysis:

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1. Heterogeneous catalysis with gaseous reactants:

(a) $2SO_2(g) + O_2(g) \xrightarrow{Pt(s)/V_2O_5(s)} 2SO_3(g)$

(b) $N_2(g) + 3H_2(g) \xrightarrow{Fe(s)} 2NH_3(g)$

(c) $H_2C=CH_2 + H_2 \xrightarrow{Ni(s)} CH_3 - CH_3$

above these reactions Pt, Fe, Ni are finely divided.

2. Heterogeneous catalysis with liquid reactants:

(a) $2H_2O_2(l) \xrightarrow{MnO_2(s)} 2H_2O(l) + O_2(g)$

(b) $C_6H_6(l) + CH_3COCl(l) \xrightarrow{\text{anhydrous } AlCl_3} C_6H_5COCH_3 + HCl$

3. Heterogeneous catalysis with solid reactants:

(a) $2KClO_3(s) \xrightarrow[1]{MnO_2} 2KCl + 3O_2$

(X) 2021 Derive expression for kinetics of unimolecular Heterogeneous surface reactions?

⇒ If there is a ~~single~~ single reactant, A and S is the substrate, the elementary processes may be depicted as follows:

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$$A + S \underset{K_{-1}}{\overset{K_1}{\rightleftharpoons}} AS$$

$$AS \xrightarrow{K_2} \text{products}$$

Suppose θ be the fraction of the surface covered by A at any instant t and pressure P. A. to Langmuir-Hinshelwood the reaction rate would be,

$$\therefore \text{rate} = r = K_2\theta \quad \text{---} \quad (1)$$

If we assume a steady-state approximation for [AS], we get;

$$d[AS]/dt = K_1[A][S] - K_{-1}[AS] - K_2[AS] = 0 \quad \text{----} \quad (2)$$

$$\Rightarrow [AS] = K_1[A][S]/(K_{-1}+K_2) \quad \text{------} \quad (3)$$

let, C_s is the total conc. of the surface sites of the catalyst. therefore, conc. of vacant sites, $[S] = C_s(1-\theta)$ and conc. of occupied sites; $[AS] = C_s\theta$

From eqn (3),

$$C_s\theta = K_1[A]C_s(1-\theta)/(K_{-1}+K_2)$$

$$\Rightarrow C_s\theta + \frac{K_1[A]C_s\theta}{K_{-1}+K_2} = \frac{K_1[A]C_s}{K_{-1}+K_2}$$

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$$\Rightarrow \theta\left(\frac{K_{-1}+K_2+K_1[A]}{K_{-1}+K_2}\right) = \frac{K_1[A]}{K_{-1}+K_2}$$

$$\Rightarrow \theta = \frac{K_1[A]}{(K_1[A]+K_{-1}+K_2)} \quad \text{---- (4)}$$

and, $$r = \frac{K_1 K_2[A]}{(K_1[A]+K_{-1}+K_2)} \quad \text{---- (5)}$$

$$\Rightarrow \frac{1}{r} = \frac{K_1[A]+K_{-1}+K_2}{K_1 K_2[A]} = \frac{K_1[A]}{K_1 K_2[A]} + \frac{K_{-1}+K_2}{K_1 K_2[A]}$$

$$\Rightarrow \frac{1}{r} = \frac{1}{K_2} + \frac{K_{-1}+K_2}{K_1 K_2[A]} \quad \text{---- (6)}$$

For the gaseous reactions, concentration [A] can be replaced by partial pressure, P_A and we get,

$$\frac{1}{r} = \frac{1}{K_2} + \frac{K_{-1}+K_2}{K_1 K_2 P_A} \quad \text{---- (7)}$$

A plot of $1/r$ against $1/P_A$ give a straight line with slope $\left(\frac{K_{-1}+K_2}{K_1 K_2}\right)$.

case-I : If $K_2 >> K_1 P_A + K_{-1}$ then

$$r = \frac{K_1 K_2 P_A}{K_1 P_A + K_{-1} + K_2} = \frac{K_1 K_2 P_A}{K_2} = K_1 P_A \quad [\because K_2 >> K_1 P_A + K_{-1}] \quad \text{---- (8)}$$

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case-II: If $K_L << K_1P_A + K_{-1}$, then,

$$r = \frac{K_1 K_L P_A}{K_1 P_A + K_{-1} + K_L} = \frac{K_1 K_L P_A}{K_1 P_A + K_{-1}} = \frac{K_1 K_L P_A}{K_{-1}\left(\frac{K_1}{K_{-1}} P_A + 1\right)}$$

$$r = \frac{K K_L P_A}{K P_A + 1} \quad \text{---} \quad \left[\text{let, } \frac{K_1}{K_{-1}} = K\right] \quad \text{---} \quad (9)$$

Eqn (9) is identical to Langmuir adsorption isotherm. So,

(a) at low pressure, i.e. $KP_A << 1$,

$$r = K K_L P_A$$

The reaction would be first order.

(b) at high pressure, i.e. $KP_A >> 1$

$$r = K_L$$

The reaction would be zero order.

(*) 2016, 2018, 2020, 2022, 2021(t), 2022(t)
Derive expression for kinetics of heterogeneous bimolecular surface reaction? what will be the nature of kinetics when gas is slightly absorbed and when strongly absorbed?

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⇒ consider a reaction in which two molecules A & B react on a surface & get absorbed on neighboring sites. The process may take place in two ways:

$$A + S \underset{K_{-1}}{\overset{K_1}{\rightleftharpoons}} AS$$

$$B + S \underset{K_{-2}}{\overset{K_2}{\rightleftharpoons}} BS$$

$$AS + BS \xrightarrow{K_3} \text{products}$$

Suppose θ_1 & θ_2 be the fraction of the surface covered by adsorption of A and B respectively. The fraction of the surface which is vacant, $(1-\theta_1-\theta_2)$. The rate of formation of products is given as follows:

$$r = K_3 \theta_1 \theta_2 \quad \text{------} (1)$$

on applying steady-state approximation to [AS] and [BS], we have,

$$d[AS]/dt = K_1[A][S] - K_{-1}[AS] - K_3[AS][BS] = 0 \quad \text{----} (2)$$

$$d[BS]/dt = K_2[B][S] - K_{-2}[BS] - K_3[AS][BS] = 0 \quad \text{----} (3)$$

Let C_s is the total conc. of the surface sites of the catalyst. Therefore,

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$[AS] = C_s \theta_1$

$[BS] = C_s \theta_2$

$[S] = C_s (1 - \theta_1 - \theta_2)$

on substituting the values $[AS]$, $[BS]$ and $[S]$ in eqn ② & ③ we get,

$$K_1 [A] C_s (1 - \theta_1 - \theta_2) - K_{-1} C_s \theta_1 - K_3 C_s \theta_1 \theta_2 = 0 \quad \text{---- ④}$$

$$K_2 [B] C_s (1 - \theta_1 - \theta_2) - K_{-2} C_s \theta_2 - K_3 C_s \theta_1 \theta_2 = 0 \quad \text{---- ⑤}$$

If it is assumed that the rate determining step is the chemical reaction btwn the adsorbed molecules, then $K_3 \to 0$ and eqn ④ and ⑤ become as follows:

$$K_1 [A] (1 - \theta_1 - \theta_2) = K_{-1} \theta_1$$

$$K_2 [B] (1 - \theta_1 - \theta_2) = K_{-2} \theta_2$$

or, $$\theta_1 = \frac{K_1}{K_{-1}} [A] (1 - \theta_1 - \theta_2) = K_1 [A] (1 - \theta_1 - \theta_2) \quad \text{---- ⑥}$$

$$\theta_2 = \frac{K_2}{K_{-2}} [B] (1 - \theta_1 - \theta_2) = K_2 [B] (1 - \theta_1 - \theta_2) \quad \text{---- ⑦}$$

where, $K_1 = \frac{K_1}{K_{-1}}$ and $K_2 = \frac{K_2}{K_{-2}}$

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on solving θ_1 and θ_2, we obtain,

$$\theta_1 = K_1[A] / (1 + K_1[A] + K_2[B])$$

$$\theta_2 = K_2[B] / (1 + K_1[A] + K_2[B])$$

put the value of θ_1 and θ_2 in eqn ①, we get

$$r = K_3 K_1 K_2 [A][B] / (1 + K_1[A] + K_2[B]) \quad \text{----- ⑧}$$

when P_A and P_B donate the partial pressure of A and B for the gaseous reaction, eqn ⑧ becomes,

$$r = K_3 K_1 K_2 P_A P_B / (1 + K_1 P_A + K_2 P_B) \quad \text{----- ⑨}$$

Three special cases of important significance may arise from eqn ⑨,

case-I: If each gas (A & B) gets adsorbed very slightly, then $K_1 P_A << 1$ and $K_2 P_A << 1$, so

$$r = K_3 K_1 K_2 P_A P_B \quad \text{----- ⑩}$$

This means reaction will be second order.

case-II: When one reactant, A, more strongly adsorbed than B then $K_1 P_A >> K_2 P_B$. So,

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$$r = K_3 K_1 K_2 P_A P_B / (1 + K_1 P_A)^2 \quad \text{---(11)}$$

This means reaction will be first order with respect to B.

Case-III: when one reactant, A, very strongly adsorbed, then $K_1 P_A \gg K_2 P_B$ and $K_1 P_A \gg 1$. so,

$$r = K_3 K_1 K_2 P_A P_B / (K_1 P_A)^2$$

$$\therefore r = K_3 K_2 P_B / K_1 P_A \quad \text{---(12)}$$

This means that the rate will depend on the concentration of strongly adsorbed reactant.

2021, 2015, 2021(t)

Derive an expression for specific rate constant using absolute rate theory in heterogeneous gas reaction?

⇒ Let us consider a bimolecular reaction btwn A & B involving a reaction center, S, on the surface.

Then $A + B + S \rightleftharpoons A\text{-}B\text{-}S \longrightarrow$ Products ---(1)

In this type of reaction there exists an equilibrium btwn the reactants (A and B) and activated complex.

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If C_A, C_B and C_S are the concentration of A, B and S respectively, the value of equilibrium constant $K_\ddagger$ of the activated state is obtained by appling the law of mass action to eq. ①,

$$K_\ddagger = C_\ddagger / C_A C_B C_S \quad \text{-----②}$$

where $C_\ddagger$ is the conc. of the activated complex.

If we consider the partition function, eq. ② becomes as,

$$K_\ddagger = \theta_\ddagger / \theta_A \theta_B \theta_S \exp(-E_a/RT) \quad \text{----③}$$

where E_a is the activation energy. From ② & ③,

$$\frac{C_\ddagger}{C_A C_B C_S} = \frac{\theta_\ddagger}{\theta_A \theta_B \theta_C \exp(-E_a/RT)}$$

$$\Rightarrow C_\ddagger = \frac{C_A C_B C_S \theta_\ddagger}{\theta_A \theta_B \theta_S \exp(-E_a/RT)} \quad \text{----④}$$

The rate of bimolecular reaction,

$$r = \frac{C_\ddagger KT}{h} \quad \text{-----⑤}$$

Substituting eqn ④ in eqn ⑤ we get,

$$r = \frac{C_A C_B C_S \theta_\ddagger KT}{h \theta_A \theta_B \theta_S \exp(-E_a/RT)} \quad \text{----⑥}$$

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The rate of the reaction may also be given by;

$$r = K_r C_A C_B \quad \text{----} (7)$$

where K_r is the specific rate constant. on comparing Eqn (6) and (7) we get,

$$K_r = \frac{C_s \theta_{\neq} KT}{h \theta_A \theta_B \theta_s \exp(-E_a/RT)} \quad \text{------} (8)$$

The s part of the surface possesses no transitional, no rotational but it has only vibrational contribution is of the order of unity. Thus $\theta_{\neq}/\theta_s$ may be taken as unity (means 1). Hence eqn (8) becomes,

$$K_r = \frac{C_s KT}{h \theta_A \theta_B \exp(-E_a/RT)} \quad \text{----} (9)$$

which is the desired expression of specific rate constant.

(X) 2018, 2021, 2017 Explain (i) catalytic poisons (ii) catalytic promoters?

⇒ (i) catalytic poisons: very often a heterogeneous catalyst in rendered ineffective by the presence

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of small amount of impurities in the reactants. The substance which destroys the activity of the catalyst to accelerate a reaction, is called catalytic poisons and the process is called catalytic poisoning. EX.

(1) The pt used in the oxidation of SO_2, is poisoned by AS_2O_3.

$$2SO_2 + O_2 \xrightarrow{Pt} 2SO_3$$

2. The Fe catalyst is poisoned by H_2S in haber process

$$N_2 + 3H_2 \xrightarrow{Fe} 2NH_3$$

3. The pt is poisoned by CO in the oxidation of hydrogen,

$$2H_2 + O_2 \xrightarrow{Pt} 2H_2O$$

⇒ (ii) Catalytic promoters: The activity of a catalyst can often be increased by addition of a small quantity of a second material. The substance which is not a catalyst but promotes the activity of a catalyst is called catalytic promoters. Ex.

1. Mo or Al_2O_3 promotes the activity of Fe catalyst in the Haber process.

$$N_2 + 3H_2 \xrightarrow{Fe} 2NH_3$$

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2018, 2017, 2022, 2022 (4)

⊛ Explain function of catalyst in term of Gibb's free energy of activation?

⇒ In general, a catalyst provides alternative path for the reaction that is having lower free energy of activation. The lowering of free energy is probably attributed to a decrease in energy of activation or a higher frequency factor or both. From the activated complex theory, we can write;

$$K_f = \frac{K_B T \exp(-\Delta G_f^\circ / RT)}{h} \quad \text{---} \quad (1)$$

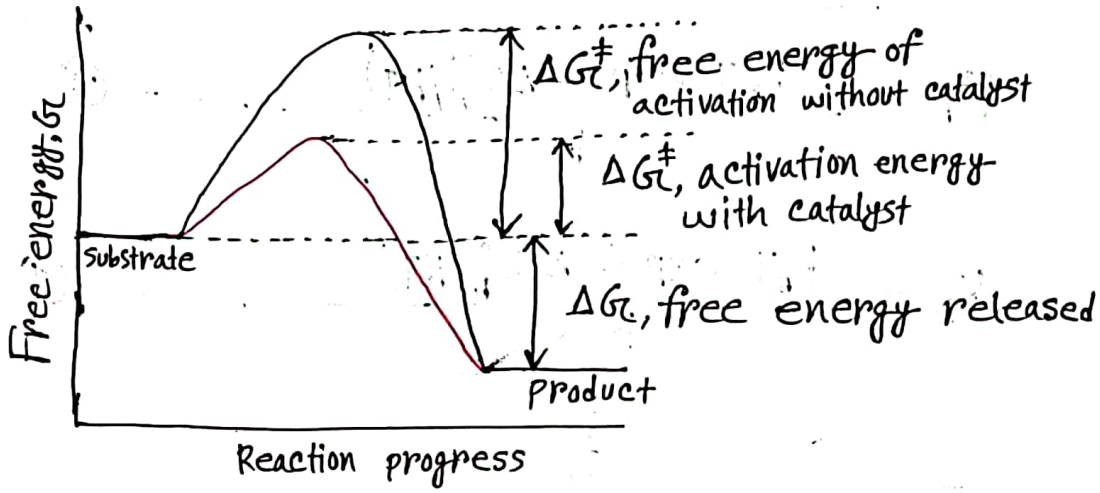
where, K_f is the rate constant for the forward reaction, K_B is the Boltzmian constant and G_f° is the standard Gibbs free energy of activation

In presence of a catalyst, it can be write,

$$K_f' = \frac{K_B T \exp(-\Delta G_f'^\circ / RT)}{h} \quad \text{---} \quad (2)$$

where (') denotes a catalyzed reaction.

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From the Figure it is evident that the free energy of activation gets lowered for both forward & backward reactions without altering the overall free energy change of the reaction. This means that a catalyst is able to change the rate of both forward & backward reactions. But ΔG°_{f} does not get changed. Therefore, the equilibrium constant K will not get change in the presence of catalyst because ΔG°_{f} is related with equilibrium constant. It further reveals that a catalyst helps in attaining the equilibrium position rapidly but does not help in changing the relative proportion of reactants & products at equilibrium. It is possible to write the rate constant for the forward and

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backward reactions as follows,

$$K_f = \frac{K_B T \exp(-\Delta G_f^{\circ}/RT)}{h} \quad \text{----③}$$

$$K_b = \frac{K_B T \exp(-\Delta G_b^{\circ}/RT)}{h} \quad \text{----④}$$

Now, ③ ÷ ④ we get,

$$\frac{K_f}{K_b} = \frac{\exp(-\Delta G_f^{\circ}/RT)}{\exp(-\Delta G_b^{\circ}/RT)} = \exp\left[(-\Delta G_f^{\circ}/RT) + (\Delta G_b^{\circ}/RT)\right]$$

$$\Rightarrow K = \exp(-\Delta G^{\circ}/RT)$$

where, $K = \frac{K_f}{K_b}$; $\Delta G^{\circ} = \Delta G_f^{\circ} - \Delta G_b^{\circ}$

Since ΔG° has the same value in the presence or absence of a catalyst, K remains the same.

2021, 2021 (t)

⊛ Write the reactions btwn CO and H_2 in presence of different catalysts ?

⇒ when CO and H_2 reacts, it gives three different products due to use of three different catalysts.

(i) $CO + 3H_2 \longrightarrow CH_4 + H_2O$ (For Ni catalyst)

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(ii) $CO + H_2 \longrightarrow HCHO$ (for Cu catalyst)

(iii) $CO + 2H_2 \longrightarrow CH_3OH$ (for $ZnO + Cr_2O_3$ catalyst)

⊛ 2017 What is Homogeneous catalysis? Discuss the mechanism of homogeneous catalysis?

⇒ Homogeneous catalysis refers to the reactions where the catalyst is in the same phase as the reactants. Now discuss mechanism with some example:

Example-1:

$$2SO_2 + O_2 \xrightarrow{NO} 2SO_3$$

Mechanism:

$$2NO + O_2 \longrightarrow 2NO_2 \text{ (intermediate compound)}$$

$$NO_2 + SO_2 \longrightarrow SO_3 + NO$$

Example-2:

$$C_2H_5OH + C_2H_5OH \longrightarrow (C_2H_5)_2O + H_2O$$

Mechanism:

$$C_2H_5OH + H_2SO_4 \longrightarrow C_2H_5HSO_4 + H_2O$$

$$C_2H_5HSO_4 + C_2H_5OH \longrightarrow (C_2H_5)_2O + H_2SO_4$$

Example-3:

$$2KClO_3 \longrightarrow 2KCl + 3O_2$$

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Mechanism:

$$2KClO_3 + 6MnO_2 \longrightarrow 6MnO_3 + 2KCl$$

$$6MnO_3 \longrightarrow 6MnO_2 + 3O_2$$

Example-4:

$$C_6H_6 + CH_3Cl \longrightarrow C_6H_5CH_3 + HCl$$

Mechanism:

$$CH_3Cl + AlCl_3 \longrightarrow [CH_3][AlCl_4]$$

$$C_6H_6 + [CH_3][AlCl_4] \longrightarrow C_6H_5CH_3 + AlCl_3 + HCl$$

⊛ 2019(t)
DISCUSS the theory of homogeneous catalysis?

⇒ In general, the intermediate compound formation theory applies to homogeneous catalytic reactions. At first a highly reactive intermediate compound is formed from a reactant and catalyst then it reacts with the second reactant to yield the product, releasing the catalyst. Let the general reaction,

$$A + B \xrightarrow{C} AB$$

where C acts as catalyst. The reaction proceeds through the reactions:

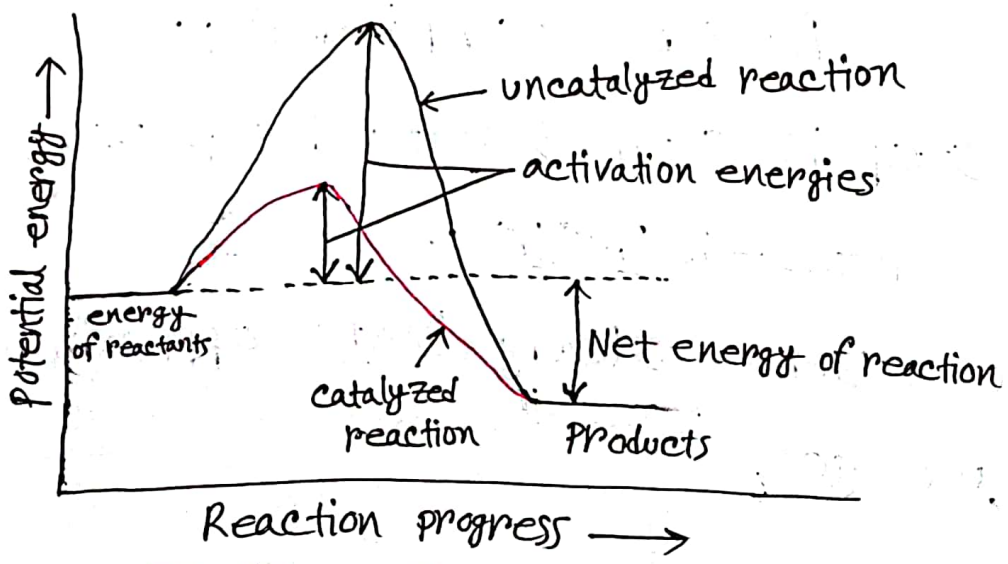
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$$A + C \longrightarrow AC \quad \text{----------} (2)$$

$$AC + B \longrightarrow AB + C \quad \text{----------} (3)$$

where AC is an intermediate. The activation energies of the reactions (2) and (3) are lower than that of the reaction (1). Hence the involvement of the catalyst in the formation of the intermediate compound and its subsequent decomposition, accelerates the rate of the reaction (1) which was originally very slow. The catalyst is regenerated in the last step and more and more product will be formed. Thus the rate of homogeneous catalysis reaction is depends on the concentration of the catalyst. It means the rate of the reaction will increase if the concentration of the catalyst is increased.



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2019(f)

⊛ Derive the rate eqn for homogeneous catalysis ~~for~~ and how the rate of such reactions related to the concentrations of catalyst?

⇒ According to homogeneous catalysis theory, the catalyst first form an intermediate compound with the reactant. This reactant with which catalyst combines is often called as substrate. For example

$$A + X \underset{K_2}{\overset{K_1}{\rightleftharpoons}} AX$$

where A is substrate and X is catalyst, AX is intermediate compound and K_1 and K_2 are the rate constant for the forward & backward reactions.

Intermediate Compound AX then reacts with reactant B to form the product.

$$AX + B \xrightarrow{K_3} AB(\text{product}) + X$$

This step is slow & rate determining step. Thus

$$\text{rate} = K_3 [AX][B] \quad \cdots\cdots (1)$$

If we assume steady-state approximation for [AX], we get,

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$$d[AX]/dt = K_1[A][X] - K_2[AX] - K_3[AX][B] = 0$$

$$\Rightarrow K_1[A][X] = K_2[AX] + K_3[AX][B]$$

$$\Rightarrow [AX] = \frac{K_1[A][X]}{K_2 + K_3[B]} \quad \text{------} (2)$$

Substituting this value in eqn (1), we get,

$$\text{rate} = \frac{K_3 K_1[A][X][B]}{K_2 + K_3[B]} \quad \text{-----} (3)$$

From eqn (3) two cases may arise:

(i) If $K_2 << K_3[B]$, then eqn (3) becomes,

$$\text{rate} = \frac{K_3 K_1[A][X][B]}{K_3[B]} = K_1[A][X] \quad \text{----} (4)$$

(ii) If $K_2 >> K_3[B]$, then eqn (3) becomes

$$\text{rate} = \frac{K_3 K_1[A][X][B]}{K_2} \quad \text{------} (5)$$

From eqn (4) & (5), it flows that the rate of reaction depends upon the concentration of catalyst although it is neither produced or consumed in the reaction.

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2020

⊛ What is the effect of pH on rate of acid catalyst reaction?

⇒ In acid catalyzed reaction, the r. rate is proportional to $[H^+]$ even in the presence of HA & A^-. So At low pH means At high conc. of $[H^+]$, the availability of H^+ to participate in the reaction increases & enhance r. rate. At high pH, conc. of H^+ is low. The reduced availability of protons slows the r. rate because catalytic effect of the acid is diminished.

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